KAPASA MAKASA UNIVERSITY

Mobile Device and Security and Forensic (CYS453)

Cyber Security

Year 4

Group 5

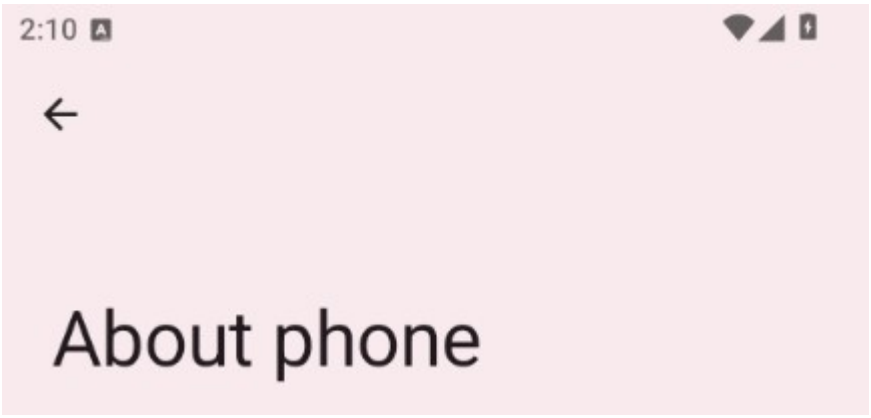
FIRST NAME	LAST NAME	S.I.N
James	Soko	20231392
Evans	Bwalya	20231947
Henry	Mutale	20231604
Bethsaida	Chileshe	20230495
Obed	Mwape	20231799
Nathan	Chintu	22175221
Salifyanji	Subakanya	20231739
Esther	Ngoma	20230442
Chimuka	Muntanga	20231270
Agripa	Mweetwa	20231822

QUESTION:

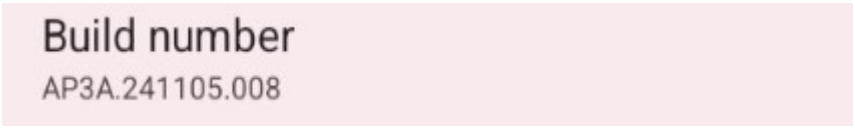
Lab 2: Android Backup Analysis – Extracting Data from Non-Rooted Devices

Enable USB Debugging on the Device

1 Navigate to Settings → About Phone



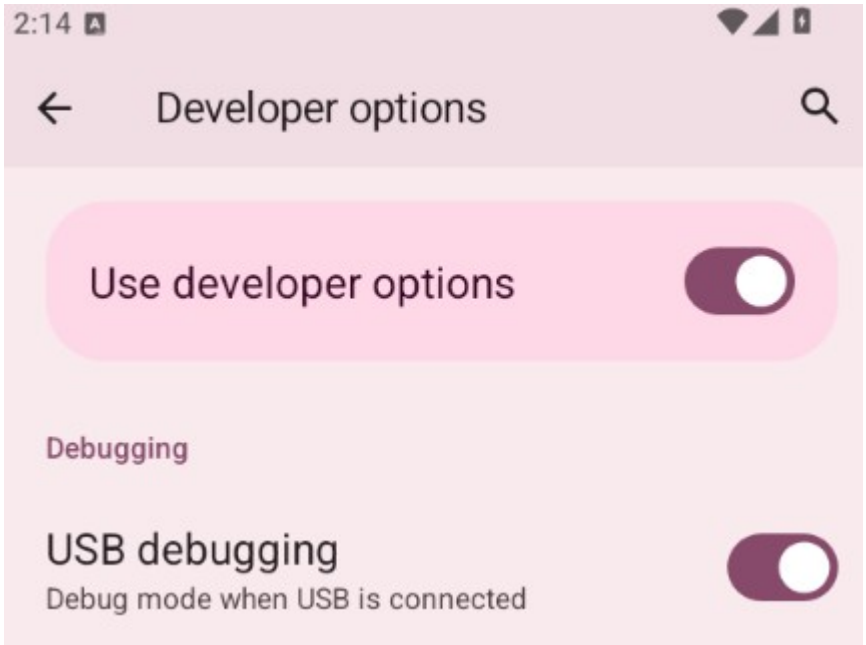
2 Tap "Build Number" 7 times to enable Developer Options



3 Go to Settings → System → Developer Options



4 Enable "USB Debugging"

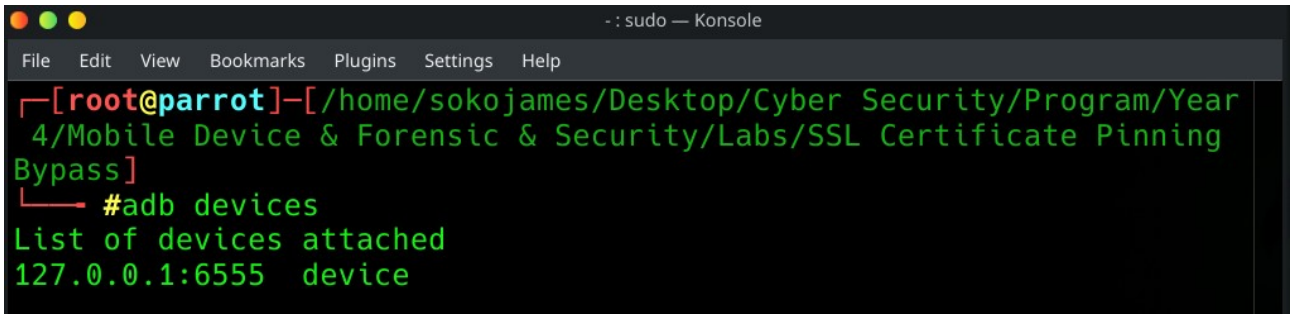


5 Connect the device and accept the RSA fingerprint prompt

Investigation Procedure

Part A: Create a Forensic Backup

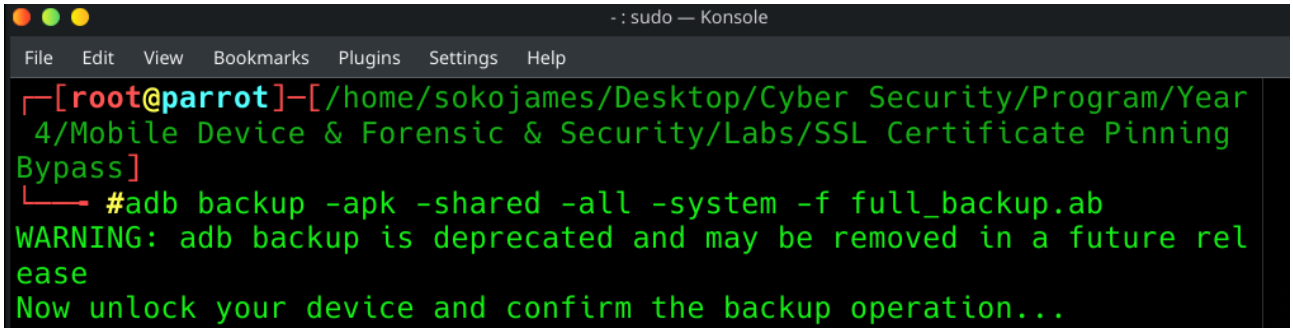
Verify device connection



First, we identify the name of our target application, in this case, its the messaging app.



Or, we can backup everything



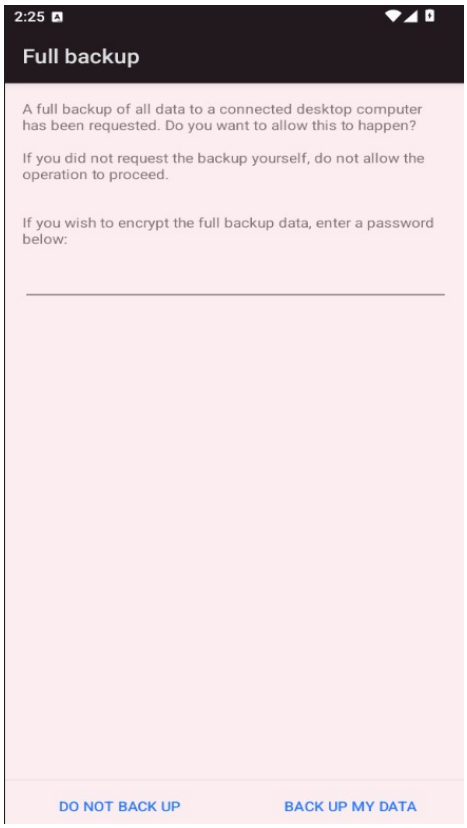
Create a backup of the target application

```
┌─[root@parrot]─[/home/sokojames/Desktop/Cyber Security/Program/Year
 4/Mobile Device & Forensic & Security/Labs/SSL Certificate Pinning
Bypass]
└──╼ #adb backup -apk -f gallery.ab com.android.gallery3d
WARNING: adb backup is deprecated and may be removed in a future rel
ease
Now unlock your device and confirm the backup operation...
```

We can now verify the backup

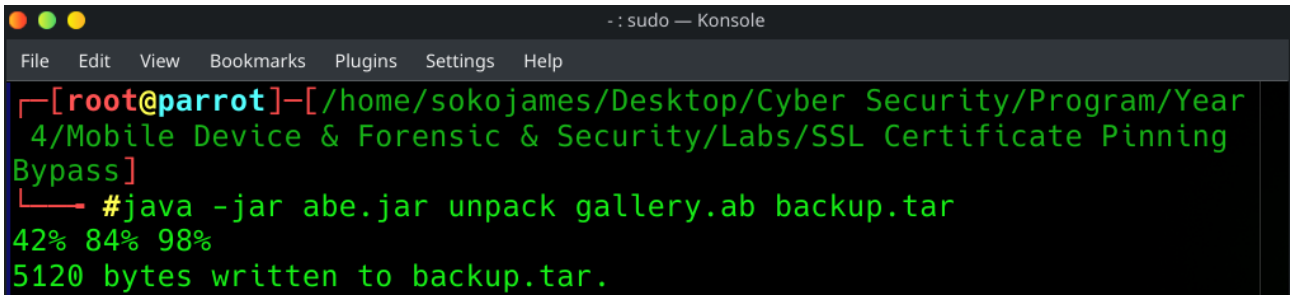
```
┌─[root@parrot]─[/home/sokojames/Desktop/Cyber Security/Program/Year
 4/Mobile Device & Forensic & Security/Labs/SSL Certificate Pinning
Bypass]
└──╼ #file gallery.ab
gallery.ab: Android Backup, version 5, Compressed, Not-Encrypted
```

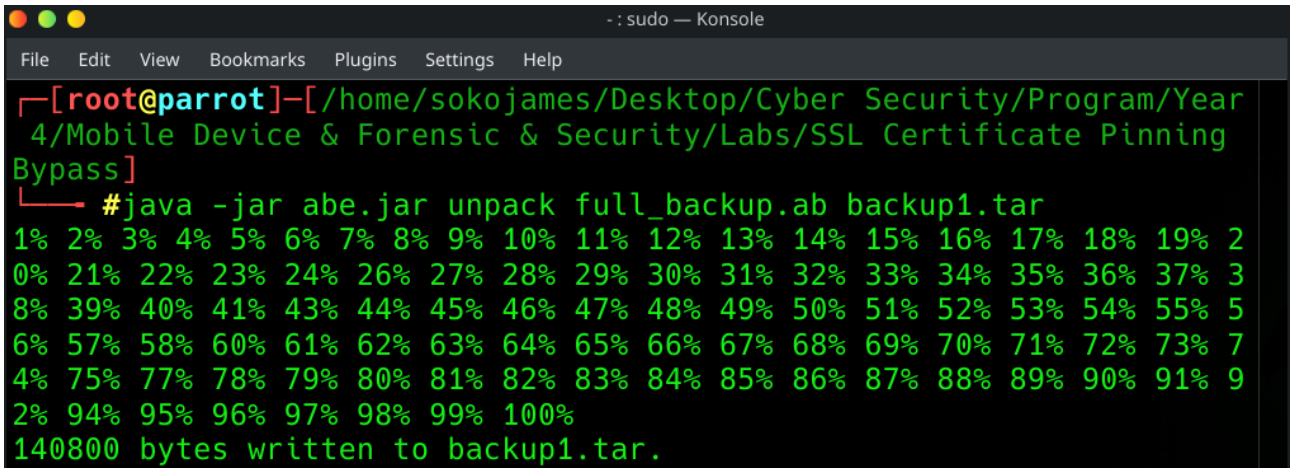
User must confirm backup on the device screen



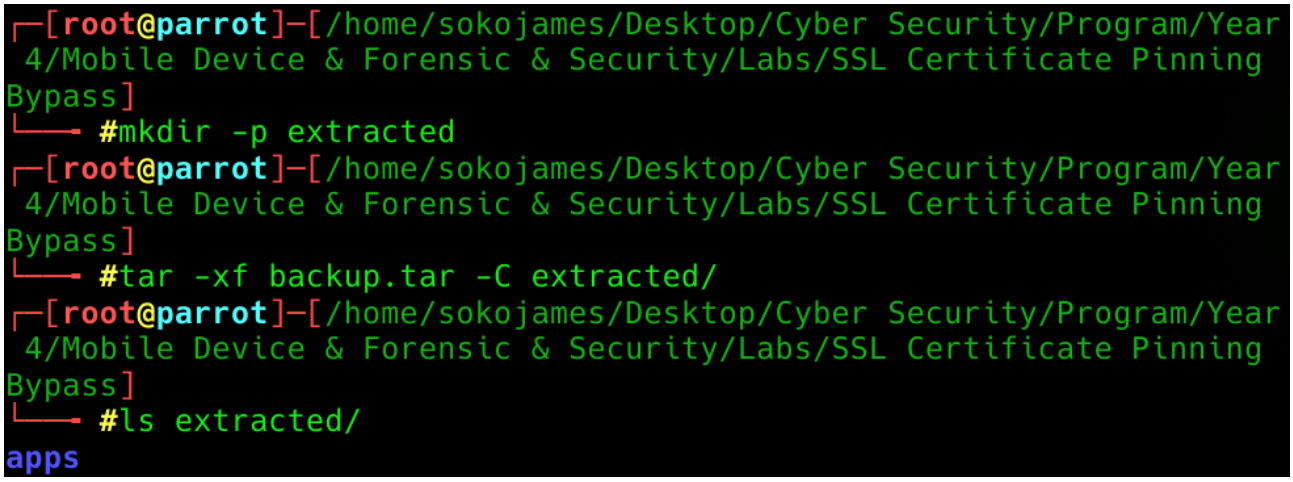
Part B: Extract the Backup File

Convert .ab to .tar format



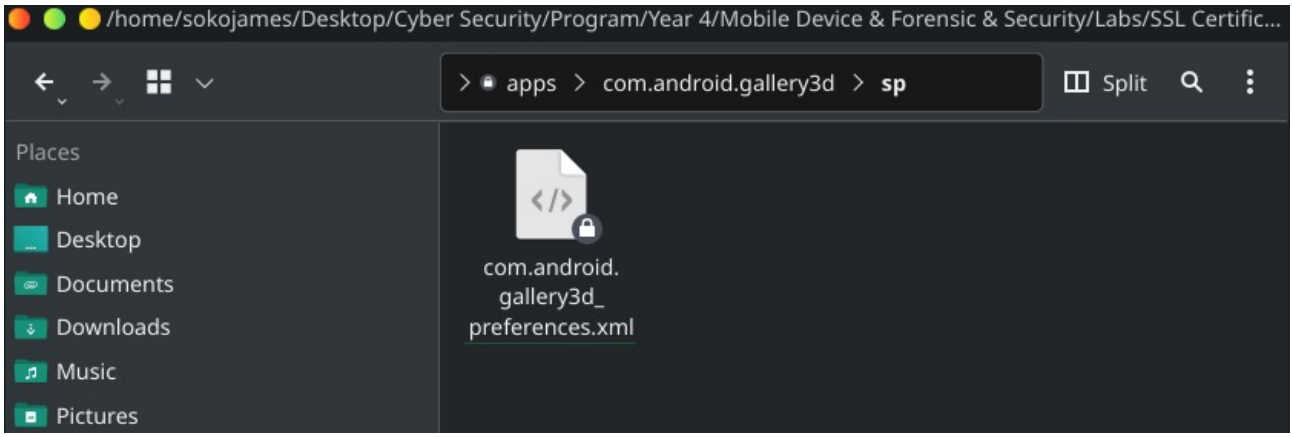


Extract the tar file

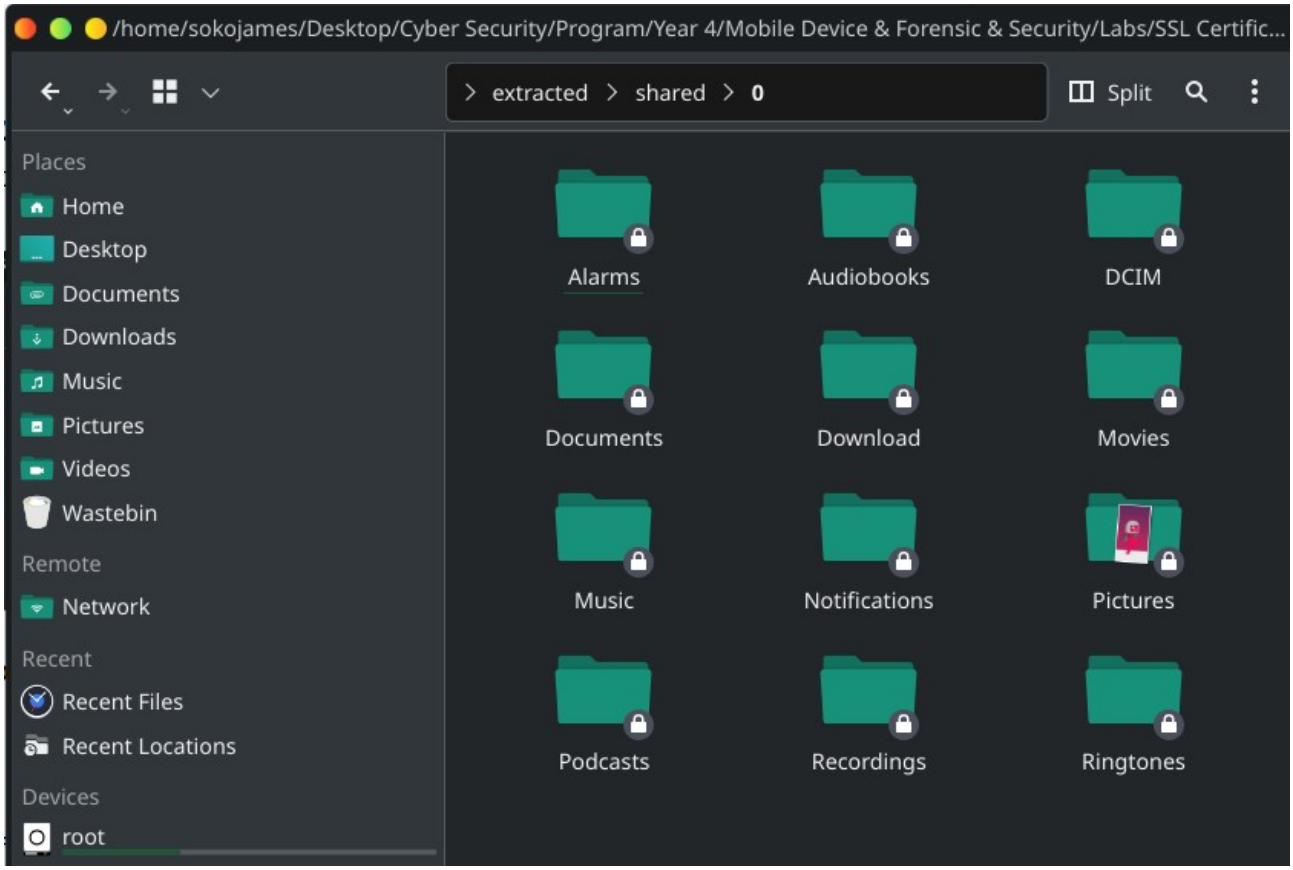


Part C: Analyze Extracted Data

1 Navigate to the extracted directory structure



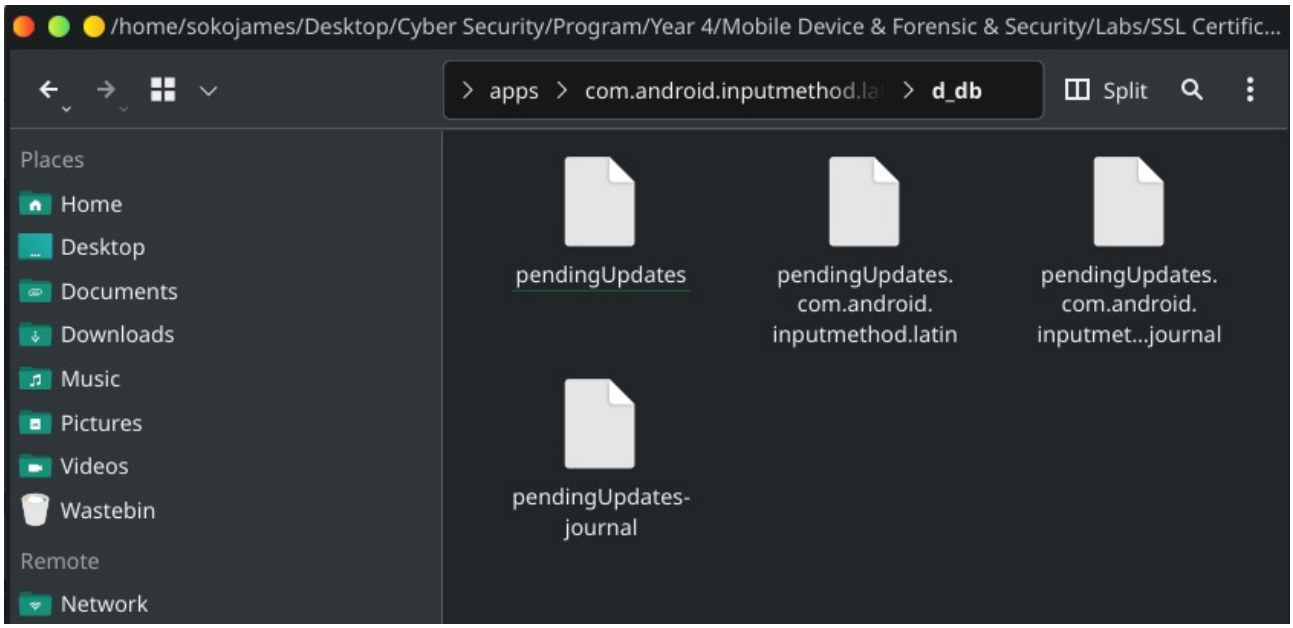
From our full backup, we have



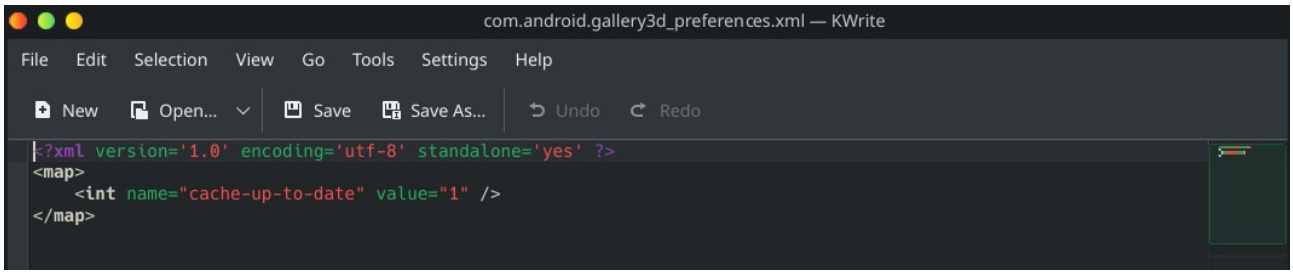
2 Examine the following key locations: apps/[package_name]/sp/ (shared preferences, XML files);

apps/[package_name]/db/ (SQLite databases); apps/[package_name]/f/ (internal storage files);

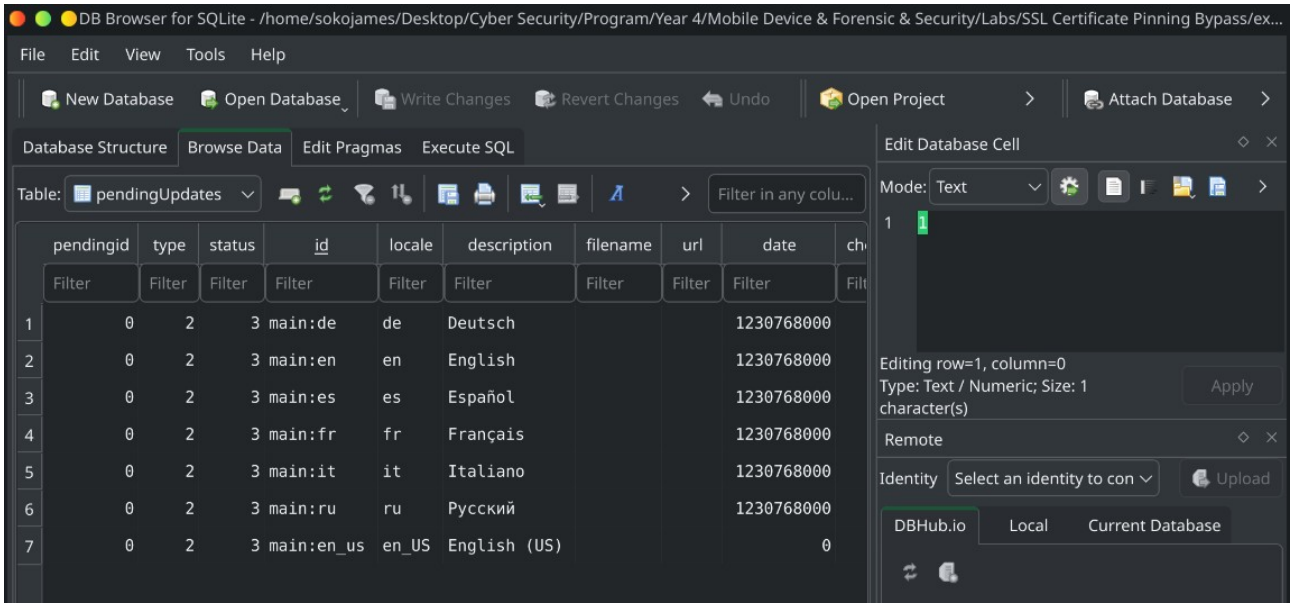
apps/[package_name]/manifest/ (AndroidManifest.xml)



3 Open any shared preferences XML files in a text editor



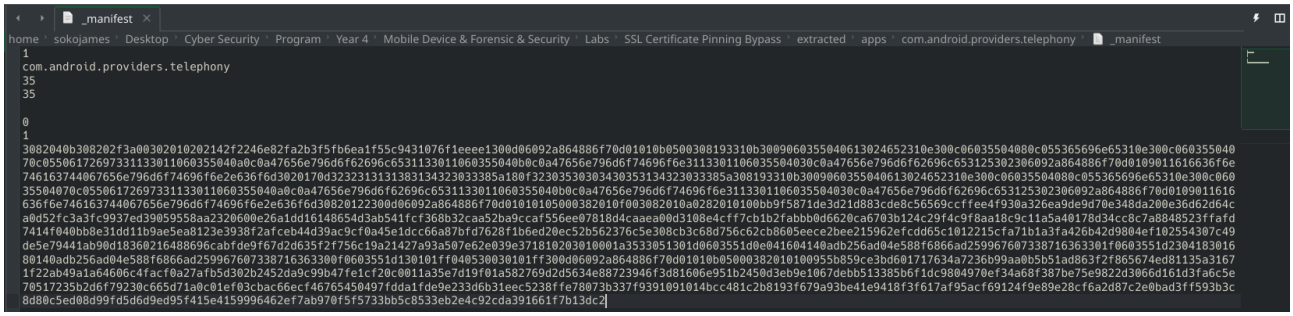
4 Open SQLite databases in DB Browser for SQLite

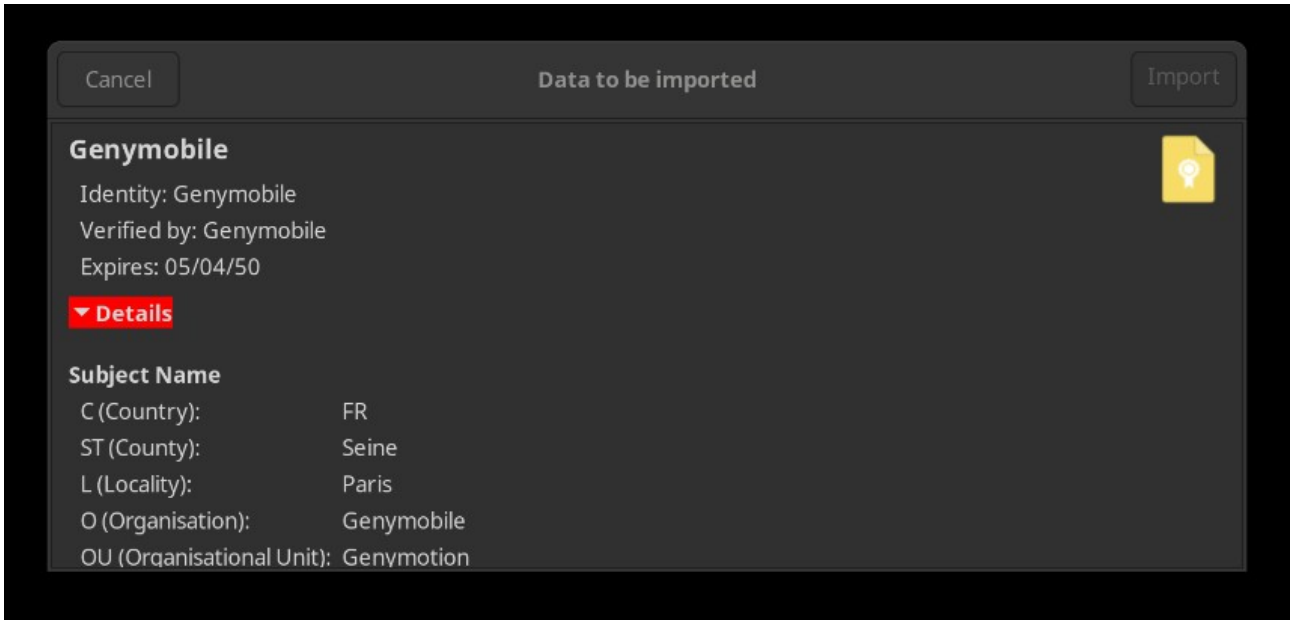


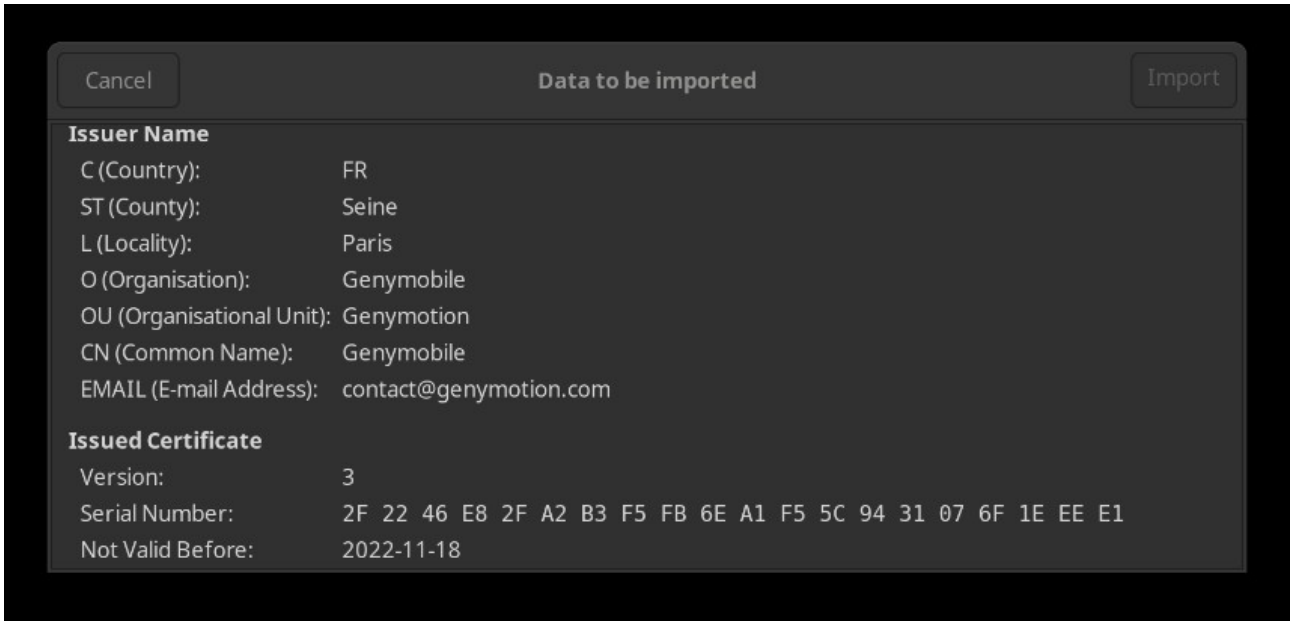
Part D: Security Assessment

Using the extracted data, identify the following:

- Sensitive data stored in plaintext (passwords, tokens, PII)







- Insecure data storage practices

